Лабораторна №3

Прикладне програмне забезпечення навчання та тестування нейронної мережі для класифікації тестових даних. Аналіз впливу архітектури мережі та параметрів навчання на точність класифікації.

Роботу виконав:

Грунда Ярослав

Студент 3-го курсу

Групи ФІ-21

Опис набору данних

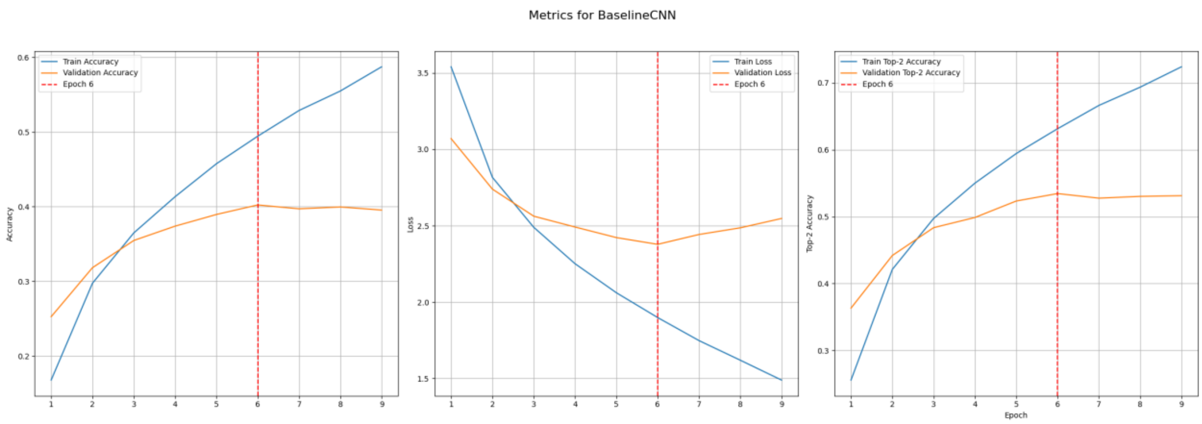
Датасет: CIFAR-100

100 класів з 600 кольорових фото 32x32 в кожному. 500 тренувальних фото і 100 тестувальних фото на клас.

Моделі

Базова модель

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 64, 32, 32]           1,792
         MaxPool2d-2           [-1, 64, 16, 16]               0
            Conv2d-3          [-1, 128, 16, 16]          73,856
         MaxPool2d-4            [-1, 128, 8, 8]               0
            Conv2d-5            [-1, 128, 8, 8]         147,584
         MaxPool2d-6            [-1, 128, 4, 4]               0
            Linear-7                  [-1, 256]         524,544
           Dropout-8                  [-1, 256]               0
            Linear-9                  [-1, 100]          25,700
================================================================
Total params: 773,476
Trainable params: 773,476
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.01
Forward/backward pass size (MB): 1.02
Params size (MB): 2.95
Estimated Total Size (MB): 3.98
----------------------------------------------------------------
```



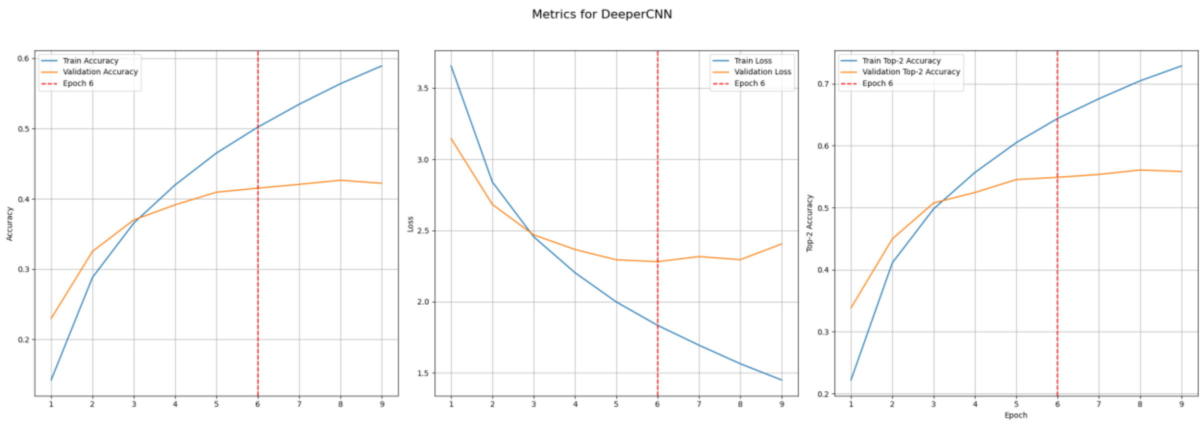

```
Epoch 6 /// acc: 0.4944 /// f1: 0.4884 /// loss: 1.8995 /// val_acc: 0.4023 /// val_f1: 0.3956 /// val_loss: 2.3785 /// time: 21.51s
Epoch 7 /// acc: 0.5288 /// f1: 0.5237 /// loss: 1.7488 /// val_acc: 0.3971 /// val_f1: 0.3916 /// val_loss: 2.4427 /// time: 21.54s
Epoch 8 /// acc: 0.5549 /// f1: 0.5503 /// loss: 1.6204 /// val_acc: 0.3996 /// val_f1: 0.3977 /// val_loss: 2.4865 /// time: 19.01s
Epoch 9 /// acc: 0.5873 /// f1: 0.5835 /// loss: 1.4902 /// val_acc: 0.3954 /// val_f1: 0.3915 /// val_loss: 2.5478 /// time: 18.70s
🔴 Early stopping triggered at epoch 9
```

Точніть на валідаційних даних 39.54%

Модель з трьома conv шарами

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 64, 32, 32]           1,792
         MaxPool2d-2           [-1, 64, 16, 16]               0
            Conv2d-3          [-1, 128, 16, 16]          73,856
         MaxPool2d-4            [-1, 128, 8, 8]               0
            Conv2d-5            [-1, 128, 8, 8]         147,584
         MaxPool2d-6            [-1, 128, 4, 4]               0
            Linear-7                  [-1, 256]         524,544
            Linear-8                  [-1, 100]          25,700
================================================================
Total params: 773,476
Trainable params: 773,476
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.01
Forward/backward pass size (MB): 1.02
Params size (MB): 2.95
Estimated Total Size (MB): 3.98
----------------------------------------------------------------
```

```
Epoch 6 /// acc: 0.5020 /// f1: 0.4968 /// loss: 1.8339 /// val_acc: 0.4152 /// val_f1: 0.4071 /// val_loss: 2.2809 /// time: 24.27s
Epoch 7 /// acc: 0.5346 /// f1: 0.5299 /// loss: 1.6949 /// val_acc: 0.4206 /// val_f1: 0.4169 /// val_loss: 2.3171 /// time: 23.69s
Epoch 8 /// acc: 0.5639 /// f1: 0.5602 /// loss: 1.5645 /// val_acc: 0.4265 /// val_f1: 0.4202 /// val_loss: 2.2951 /// time: 23.68s
Epoch 9 /// acc: 0.5892 /// f1: 0.5860 /// loss: 1.4504 /// val_acc: 0.4222 /// val_f1: 0.4172 /// val_loss: 2.4051 /// time: 23.71s
🔴 Early stopping triggered at epoch 9
```

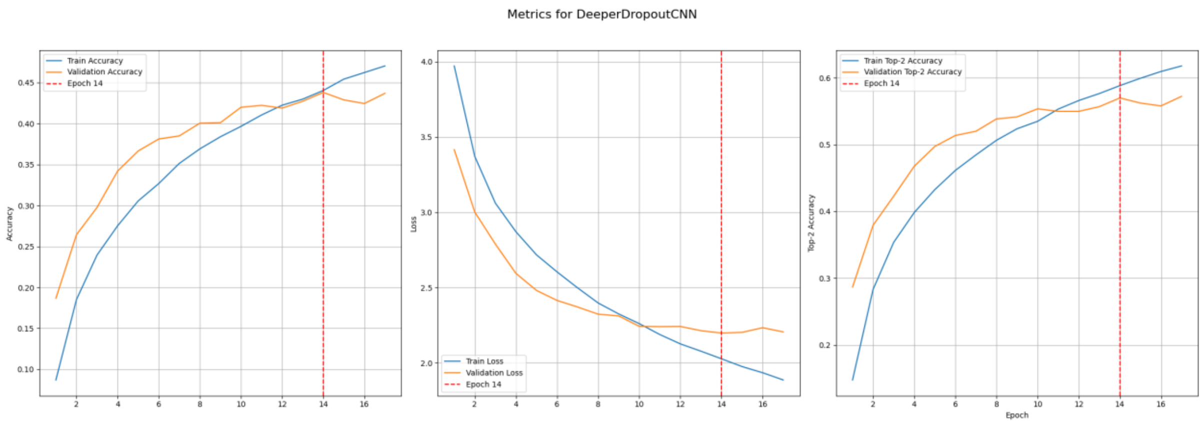


Точніть на валідаційних даних 42.22%

Модель з dropout

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 64, 32, 32]           1,792
         MaxPool2d-2           [-1, 64, 16, 16]               0
            Conv2d-3          [-1, 128, 16, 16]          73,856
         MaxPool2d-4            [-1, 128, 8, 8]               0
            Conv2d-5            [-1, 128, 8, 8]         147,584
         MaxPool2d-6            [-1, 128, 4, 4]               0
            Linear-7                  [-1, 256]         524,544
           Dropout-8                  [-1, 256]               0
            Linear-9                  [-1, 100]          25,700
================================================================
Total params: 773,476
Trainable params: 773,476
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.01
Forward/backward pass size (MB): 1.02
Params size (MB): 2.95
Estimated Total Size (MB): 3.98
```

```
Epoch 14 /// acc: 0.4403 /// f1: 0.4329 /// loss: 2.0276 /// val_acc: 0.4380 /// val_f1: 0.4327 /// val_loss: 2.1986 /// time: 23.89s
Epoch 15 /// acc: 0.4542 /// f1: 0.4467 /// loss: 1.9760 /// val_acc: 0.4289 /// val_f1: 0.4247 /// val_loss: 2.2033 /// time: 23.78s
Epoch 16 /// acc: 0.4623 /// f1: 0.4550 /// loss: 1.9351 /// val_acc: 0.4245 /// val_f1: 0.4253 /// val_loss: 2.2340 /// time: 24.38s
Epoch 17 /// acc: 0.4703 /// f1: 0.4644 /// loss: 1.8870 /// val_acc: 0.4369 /// val_f1: 0.4327 /// val_loss: 2.2063 /// time: 23.71s
🔴 Early stopping triggered at epoch 17
```

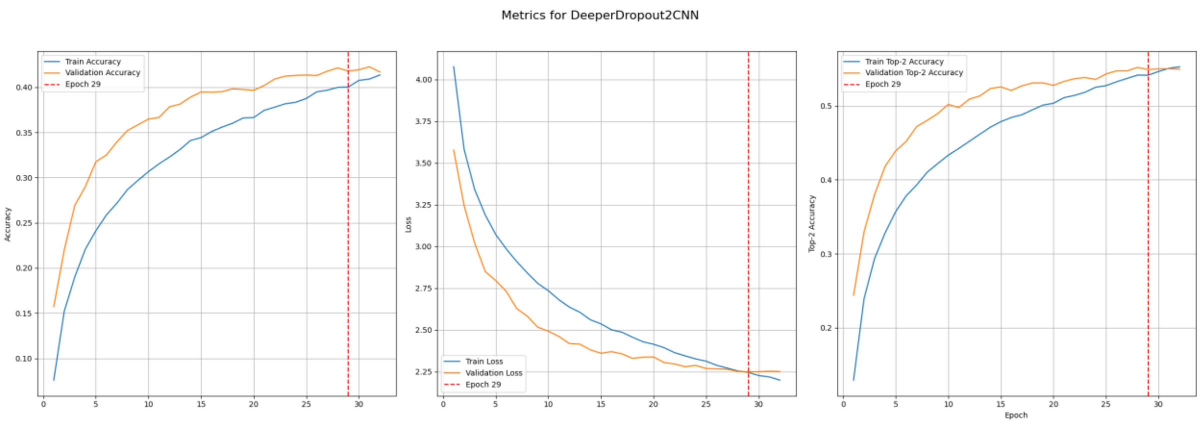


Точніть на валідаційних даних 43.69%

Модель з двома dropout

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 64, 32, 32]           1,792
         MaxPool2d-2           [-1, 64, 16, 16]               0
         Dropout2d-3           [-1, 64, 16, 16]               0
            Conv2d-4          [-1, 128, 16, 16]          73,856
         MaxPool2d-5            [-1, 128, 8, 8]               0
         Dropout2d-6            [-1, 128, 8, 8]               0
            Conv2d-7            [-1, 128, 8, 8]         147,584
         MaxPool2d-8            [-1, 128, 4, 4]               0
            Linear-9                  [-1, 256]         524,544
          Dropout-10                  [-1, 256]               0
           Linear-11                  [-1, 100]          25,700
================================================================
Total params: 773,476
Trainable params: 773,476
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.01
Forward/backward pass size (MB): 1.21
Params size (MB): 2.95
Estimated Total Size (MB): 4.17
----------------------------------------------------------------
```

```
Epoch 29 /// acc: 0.4002 /// f1: 0.3932 /// loss: 2.2452 /// val_acc: 0.4179 /// val_f1: 0.4100 /// val_loss: 2.2489 /// time: 24.97s
Epoch 30 /// acc: 0.4075 /// f1: 0.3997 /// loss: 2.2262 /// val_acc: 0.4193 /// val_f1: 0.4145 /// val_loss: 2.2496 /// time: 23.45s
Epoch 31 /// acc: 0.4090 /// f1: 0.4013 /// loss: 2.2179 /// val_acc: 0.4226 /// val_f1: 0.4163 /// val_loss: 2.2521 /// time: 23.47s
Epoch 32 /// acc: 0.4135 /// f1: 0.4067 /// loss: 2.1989 /// val_acc: 0.4169 /// val_f1: 0.4116 /// val_loss: 2.2506 /// time: 23.68s
🔴 Early stopping triggered at epoch 32
```

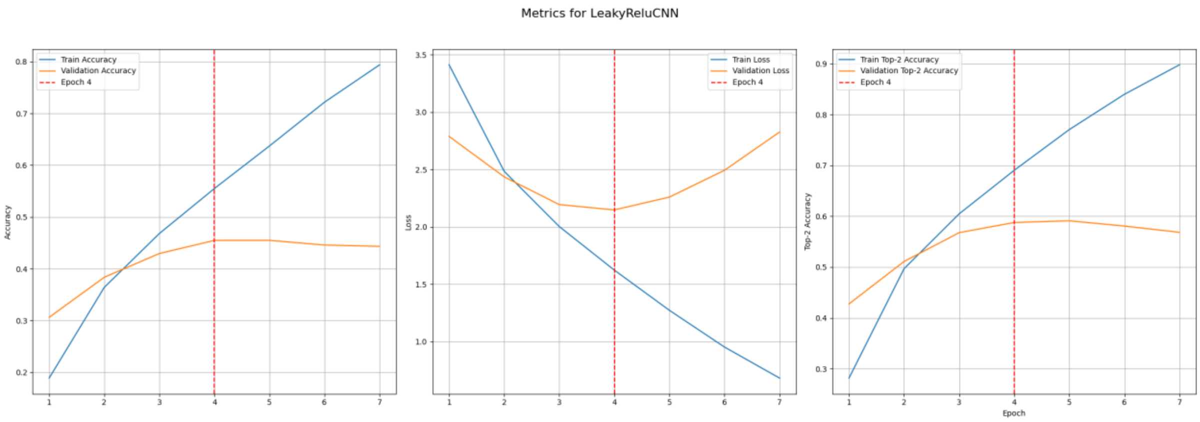


Точність на валідаційних даних 41.69%

Модель з LeakyRelu

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 64, 32, 32]           1,792
         MaxPool2d-2           [-1, 64, 16, 16]               0
            Conv2d-3          [-1, 128, 16, 16]          73,856
         MaxPool2d-4            [-1, 128, 8, 8]               0
            Conv2d-5            [-1, 128, 8, 8]         147,584
         MaxPool2d-6            [-1, 128, 4, 4]               0
            Linear-7                  [-1, 256]         524,544
            Linear-8                  [-1, 100]          25,700
================================================================
Total params: 773,476
Trainable params: 773,476
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.01
Forward/backward pass size (MB): 1.02
Params size (MB): 2.95
Estimated Total Size (MB): 3.98
----------------------------------------------------------------
```

```
Epoch 5 /// acc: 0.6372 /// f1: 0.6350 /// loss: 1.2718 /// val_acc: 0.4549 /// val_f1: 0.4487 /// val_loss: 2.2590 /// time: 23.61s
Epoch 6 /// acc: 0.7218 /// f1: 0.7205 /// loss: 0.9517 /// val_acc: 0.4459 /// val_f1: 0.4457 /// val_loss: 2.4925 /// time: 23.11s
Epoch 7 /// acc: 0.7940 /// f1: 0.7934 /// loss: 0.6811 /// val_acc: 0.4434 /// val_f1: 0.4408 /// val_loss: 2.8259 /// time: 23.10s
🔴 Early stopping triggered at epoch 7
```

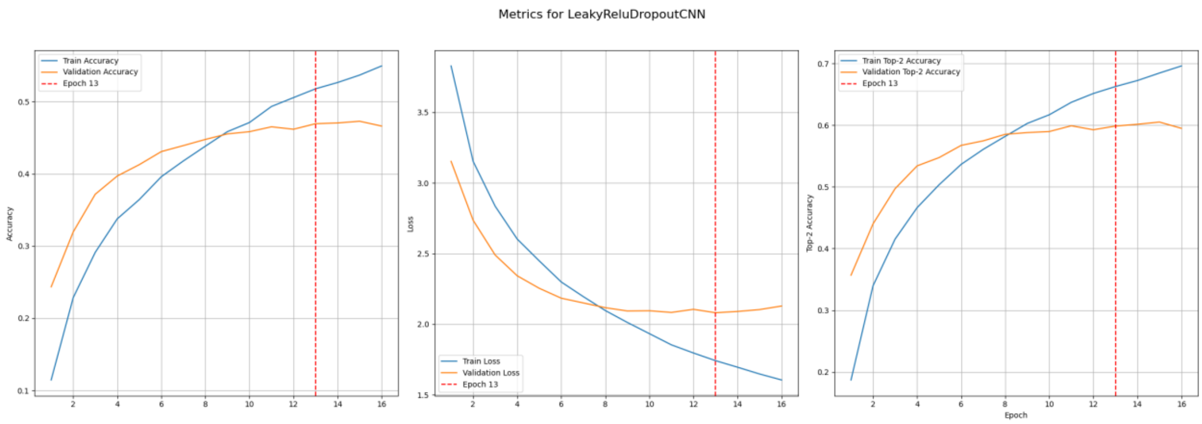


Точніть на валідаційних даних 44.34%

Модель з LeakyRelu + dropout

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 64, 32, 32]           1,792
         MaxPool2d-2           [-1, 64, 16, 16]               0
            Conv2d-3          [-1, 128, 16, 16]          73,856
         MaxPool2d-4            [-1, 128, 8, 8]               0
            Conv2d-5            [-1, 128, 8, 8]         147,584
         MaxPool2d-6            [-1, 128, 4, 4]               0
            Linear-7                  [-1, 256]         524,544
            Linear-8                  [-1, 100]          25,700
================================================================
Total params: 773,476
Trainable params: 773,476
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.01
Forward/backward pass size (MB): 1.02
Params size (MB): 2.95
Estimated Total Size (MB): 3.98
----------------------------------------------------------------
```

```
Epoch 13 /// acc: 0.5177 /// f1: 0.5137 /// loss: 1.7425 /// val_acc: 0.4694 /// val_f1: 0.4650 /// val_loss: 2.0815 /// time: 26.62s
Epoch 14 /// acc: 0.5265 /// f1: 0.5223 /// loss: 1.6960 /// val_acc: 0.4704 /// val_f1: 0.4684 /// val_loss: 2.0906 /// time: 27.08s
Epoch 15 /// acc: 0.5368 /// f1: 0.5334 /// loss: 1.6479 /// val_acc: 0.4728 /// val_f1: 0.4684 /// val_loss: 2.1039 /// time: 25.00s
Epoch 16 /// acc: 0.5494 /// f1: 0.5463 /// loss: 1.6055 /// val_acc: 0.4662 /// val_f1: 0.4637 /// val_loss: 2.1291 /// time: 25.55s
🔴 Early stopping triggered at epoch 16
```

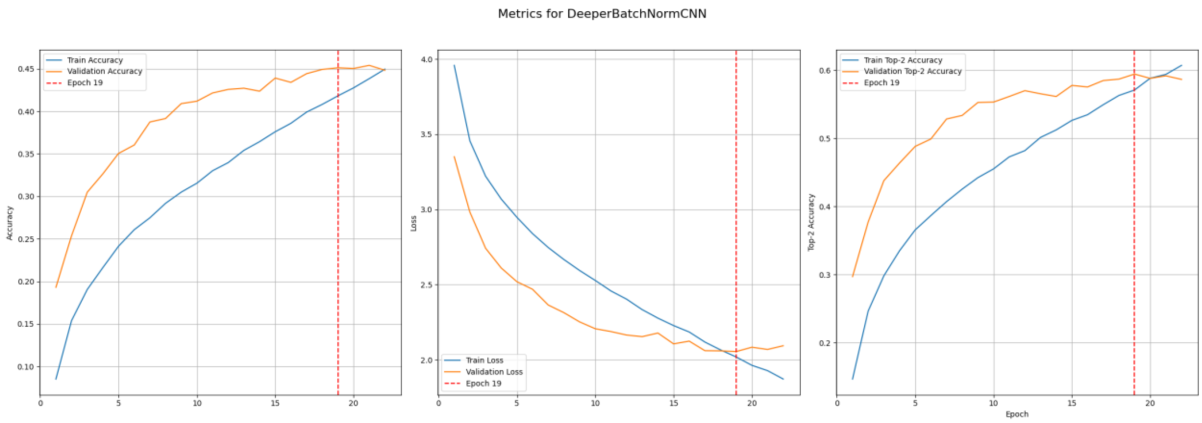


Точність на валідаційних даних 46.62%

Модель з batch normalization + dropout

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 64, 32, 32]           1,792
       BatchNorm2d-2           [-1, 64, 32, 32]             128
         MaxPool2d-3           [-1, 64, 16, 16]               0
            Conv2d-4          [-1, 128, 16, 16]          73,856
       BatchNorm2d-5          [-1, 128, 16, 16]             256
         MaxPool2d-6            [-1, 128, 8, 8]               0
            Conv2d-7            [-1, 128, 8, 8]         147,584
       BatchNorm2d-8            [-1, 128, 8, 8]             256
         MaxPool2d-9            [-1, 128, 4, 4]               0
           Linear-10                  [-1, 256]         524,544
          Dropout-11                  [-1, 256]               0
           Linear-12                  [-1, 100]          25,700
================================================================
Total params: 774,116
Trainable params: 774,116
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.01
Forward/backward pass size (MB): 1.83
Params size (MB): 2.95
Estimated Total Size (MB): 4.80
----------------------------------------------------------------
```

```
Epoch 18 /// acc: 0.4082 /// f1: 0.3974 /// loss: 2.0656 /// val_acc: 0.4492 /// val_f1: 0.4442 /// val_loss: 2.0594 /// time: 32.91s
Epoch 19 /// acc: 0.4182 /// f1: 0.4076 /// loss: 2.0188 /// val_acc: 0.4511 /// val_f1: 0.4464 /// val_loss: 2.0547 /// time: 28.81s
Epoch 20 /// acc: 0.4276 /// f1: 0.4182 /// loss: 1.9629 /// val_acc: 0.4503 /// val_f1: 0.4456 /// val_loss: 2.0835 /// time: 25.68s
Epoch 21 /// acc: 0.4384 /// f1: 0.4293 /// loss: 1.9277 /// val_acc: 0.4540 /// val_f1: 0.4499 /// val_loss: 2.0687 /// time: 25.58s
Epoch 22 /// acc: 0.4496 /// f1: 0.4405 /// loss: 1.8720 /// val_acc: 0.4482 /// val_f1: 0.4463 /// val_loss: 2.0934 /// time: 31.19s
🔴 Early stopping triggered at epoch 22
```

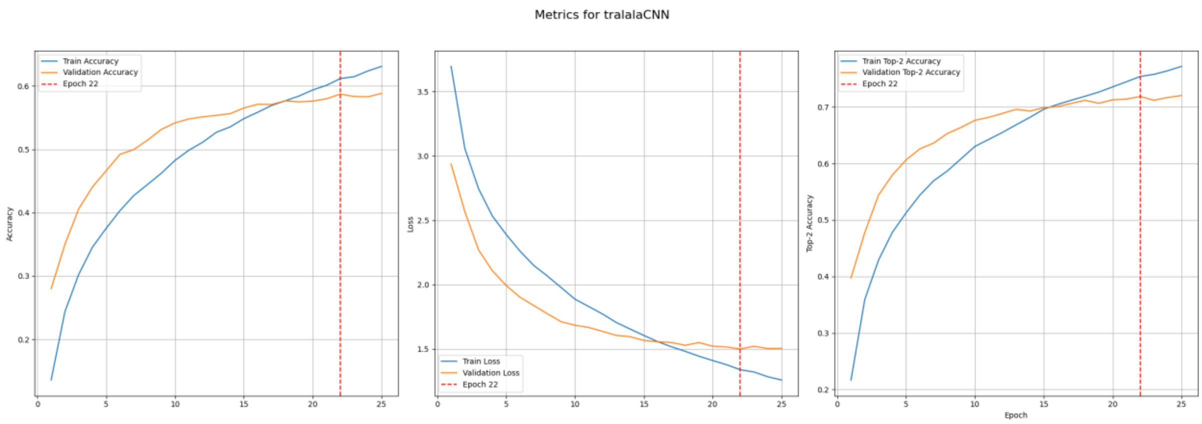


Точніть на валідаційних даних 44.82%

Модель "складна"

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 64, 32, 32]           1,792
       BatchNorm2d-2           [-1, 64, 32, 32]             128
            Conv2d-3           [-1, 64, 32, 32]          36,928
       BatchNorm2d-4           [-1, 64, 32, 32]             128
         MaxPool2d-5           [-1, 64, 16, 16]               0
         Dropout2d-6           [-1, 64, 16, 16]               0
            Conv2d-7          [-1, 128, 16, 16]          73,856
       BatchNorm2d-8          [-1, 128, 16, 16]             256
            Conv2d-9          [-1, 128, 16, 16]         147,584
      BatchNorm2d-10          [-1, 128, 16, 16]             256
        MaxPool2d-11            [-1, 128, 8, 8]               0
        Dropout2d-12            [-1, 128, 8, 8]               0
           Conv2d-13            [-1, 256, 8, 8]         295,168
      BatchNorm2d-14            [-1, 256, 8, 8]             512
        MaxPool2d-15            [-1, 256, 4, 4]               0
        Dropout2d-16            [-1, 256, 4, 4]               0
           Linear-17                  [-1, 512]       2,097,664
      BatchNorm1d-18                  [-1, 512]           1,024
          Dropout-19                  [-1, 512]               0
           Linear-20                  [-1, 100]          51,300
================================================================
Total params: 2,706,596
...
Forward/backward pass size (MB): 3.70
Params size (MB): 10.32
Estimated Total Size (MB): 14.04
```

```
Epoch 22 /// acc: 0.6119 /// f1: 0.6095 /// loss: 1.3394 /// val_acc: 0.5873 /// val_f1: 0.5842 /// val_loss: 1.5002 /// time: 54.30s
Epoch 23 /// acc: 0.6148 /// f1: 0.6123 /// loss: 1.3215 /// val_acc: 0.5835 /// val_f1: 0.5782 /// val_loss: 1.5215 /// time: 55.21s
Epoch 24 /// acc: 0.6238 /// f1: 0.6213 /// loss: 1.2843 /// val_acc: 0.5830 /// val_f1: 0.5797 /// val_loss: 1.5039 /// time: 52.00s
Epoch 25 /// acc: 0.6311 /// f1: 0.6289 /// loss: 1.2591 /// val_acc: 0.5884 /// val_f1: 0.5855 /// val_loss: 1.5051 /// time: 53.31s
🔴 Early stopping triggered at epoch 25
```



Точніть на валідаційних даних 58.84%

Табличка порівняння

	Model	Best Val_accuracy	Learning Time (s)
0	BaselineCNN	0.402269	189.552084
3	DeeperDropout2CNN	0.422572	775.506040
2	DeeperCNN	0.426453	214.434904
4	DeeperDropoutCNN	0.437998	418.787719
1	DeeperBatchNormCNN	0.454021	637.341499
5	LeakyReluCNN	0.454916	162.618952
6	LeakyReluDropoutCNN	0.472830	404.373863
7	tralalaCNN	0.588376	1276.505859

*tralalaCNN - складна (остання) модель

Як видно складна модель показала найкращу якість роботи.

Висновки

Побудовано базову CNN-модель, а також реалізовано декілька її вдосконалень з використанням:

- глибших шарів,
- нормалізації (BatchNorm),
- регуляризації (Dropout),
- альтернативних функцій активації (LeakyReLU).

Як показали експерименти чим складніша модель, тим краще вона дає результат. Також гарно показала себе функція активації LeakyRelu замість звичайної Relu.

В результаті, серед проведених експериментів, найкращою виявилась модель із глибокою архітектурою, яка включала п’ять згорткових блоків із шарами BatchNorm і Dropout, а також два повнозв’язні шари. Завдяки поєднанню нормалізації, регуляризації та поступового збільшення кількості фільтрів (від 64 до 256), модель забезпечила найвищу точність класифікації на валідаційному наборі. Точніть на валідаційних даних 58.84%.